1. Parameter Estimation Using Maximum Likelihood Estimation (MLE)

2.1 What is MLE?

- MLE is a method to estimate the parameters of a statistical model that maximize the probability (likelihood) of observing the given data.

- It is widely used because of its desirable properties like consistency and efficiency.

2.2 MLE for Normal Distribution

- For data assumed to come from a **normal distribution**, the MLE estimates for the mean (μ) and variance (σ^2) are:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2$$

Note: Unlike the unbiased sample variance formula (dividing by n−1n-1), MLE divides by n.

2.3 Practical Example: Estimating μ and σ²

Given data points:

$$x = \{4.5, 5.0, 6.2, 5.8, 4.9, 5.5, 6.0, 4.7, 5.3, 5.6\}$$

- **Step 1:** Calculate the sample mean (μ):

$$\hat{\mu} = \frac{4.5 + 5.0 + 6.2 + 5.8 + 4.9 + 5.5 + 6.0 + 4.7 + 5.3 + 5.6}{10}$$
$$= \frac{53.5}{10} = 5.35$$

- **Step 2:** Calculate the sample variance (σ²):

$$\hat{\sigma}^2 = \frac{1}{10}\sum_{i=1}^{10}(x_i - 5.35)^2$$

Calculate squared deviations:

x_i	$x_i - 5.35$	$(x_i - 5.35)^2$
4.5	-0.85	0.7225
5.0	-0.35	0.1225
6.2	0.85	0.7225
5.8	0.45	0.2025
4.9	-0.45	0.2025
5.5	0.15	0.0225
6.0	0.65	0.4225
4.7	-0.65	0.4225
5.3	-0.05	0.0025
5.6	0.25	0.0625

Sum of squared deviations:

$$0.7225 + 0.1225 + 0.7225 + 0.2025 + 0.2025 + 0.0225 + 0.4225 + 0.4225 + 0.0025 + 0.0625 = 2.9045$$

So,

$$\hat{\sigma}^2 = \frac{2.9045}{10} = 0.29045$$

What is the Method of Moments (MoM)?

I. Definition

The **Method of Moments (MoM)** is a **parameter estimation technique** in statistics. It estimates the unknown parameters of a probability distribution by **equating the theoretical moments** (from the probability model) to the **sample moments** (from the data).

$$E[X] = \frac{1}{\lambda}$$

The company tests 4 fans and records their lifespans (in hours):

$$x = \{1000, 1200, 800, 1000\}$$

Step 1: Compute the sample mean (1st sample moment):

$$\bar{x} = \frac{1000 + 1200 + 800 + 1000}{4} = \frac{4000}{4} = 1000$$

Step 2: Set sample moment = theoretical moment:

$$\bar{x} = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{\bar{x}} = \frac{1}{1000} = 0.001$$

✅ Using the **Method of Moments**, the estimated rate of failure is:

$$\hat{\lambda} = 0.001 \text{ failures per hour}$$

Example: Estimating the Rate of an Exponential Distribution

Suppose a company wants to estimate how long their electric fans last before failure. They assume that the fan lifespans follow an **exponential distribution**, which is commonly used to model time until an event (like failure).

The exponential distribution has one parameter:

λ=rate of failure\lambda = \text{rate of failure}λ=rate of failure

Its **theoretical mean** is:

E[X]=1λE[X] = \frac{1}{\lambda}

II. What Are "Moments"?

In statistics, a **moment** is a quantitative measure related to the shape of a distribution.

- **1st moment** = Mean

- **2nd moment** = Variance (technically, $E[X^2]$ minus the mean squared)
- **3rd moment** = Skewness
- **4th moment** = Kurtosis

In MoM, we usually use the first few moments (like mean and variance).

III. How Method of Moments Works

Step-by-Step Procedure:

1. **Compute sample moments** from the data.
2. **Write theoretical expressions** for those same moments in terms of the parameters.
3. **Set sample moments equal to theoretical moments.**
4. **Solve the resulting equations** to estimate the parameters.

Feature	Method of Moments (MoM)	Maximum Likelihood Estimation (MLE)
Based on	Matching moments	Maximizing likelihood
Simpler to compute	Often yes	Can be complex
Efficiency	Less efficient	More efficient
Bias	Can be biased	Often asymptotically unbiased
Universality	May fail if moments don’t exist	More flexible for a wide class of models